Zachary Cohen

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Clearance information available upon request

Systems Engineering · Autonomous & Robotic Systems · Electromechanical Integration

SUMMARY

Systems engineer with a hands-on background in electromechanical integration, mechanical design, custom PCB development, and embedded controls. Experienced working across mechanical, electrical, controls, firmware, and manufacturing boundaries, carrying field-deployable autonomous platforms from concept through live-water test. Published co-author (IEEE/RSJ IROS, ICUAS), building the hardware and running the experimental testing behind the data.

TECHNICAL SKILLS

Systems Architecture & Integration: Cross-domain electromechanical integration; interface definition & ownership in ambiguous system boundaries; integration-risk identification & mitigation; requirements traceability; verification & validation (V&V); configuration management

Electrical / PCB: Multi-layer PCB design (Altium, KiCad, Fusion Electronics); power distribution & regulation; EMI mitigation; connector selection; signal integrity; electronics bring-up

Embedded / Controls: STM32, ESP32, RP2040, ARM Cortex, ATtiny85; C/C++, MicroPython; CAN, RS-485 / Modbus RTU, I²C, SPI, UART; motor control; ABZ encoders; sensor & actuator integration

Mechanical / CAD: SolidWorks (CSWP), Fusion 360; GD&T & tolerance stack-ups; enclosure architecture for harsh environments; electromechanical packaging; thermal considerations; sealing (O-ring / U-cup); mechanical–electrical interface coordination

Prototyping, Fabrication & DFM: Additive (FDM/SLA) & subtractive (CNC) manufacturing; design for manufacturability & assembly (DFM/DFA); rapid hardware development; prototype-to-field transition; lab & test infrastructure; technical data packages (TDPs)

EXPERIENCE

Senior Systems Engineer | Johns Hopkins Applied Physics Laboratory | Laurel, MD

November 2026 – Present

Senior Electromechanical Systems Engineer | BlackSea Technologies | Baltimore, MD

2025 – 2026

- Served as cross-portfolio electromechanical systems integrator — reviewing designs across mechanical, electrical, controls, firmware, and manufacturing boundaries, and evaluating new projects at concept and mid-development for electromechanical feasibility, informing staffing and resourcing decisions.
- Led all integration on an experimental autonomous hydrofoil under a 2-month sprint, delivering on schedule and passing sea trials.
- Contributed to integration of a payload-release add-on for an autonomous surface vessel with a one-quarter target timeline.
- Identified an oversized power-distribution unit, GPIO margin shortfalls, and reliability-tier mismatches during a controls design review, driving correction before fabrication.
- Made and defended cross-domain design decisions — release mechanisms, sealing architecture, connector selection, power distribution — under schedule and procurement constraints.
- Developed enclosure, interface, and documentation standards intended to reduce ambiguity and improve repeatability across rapidly developed autonomous maritime hardware.
- Translated incomplete and evolving customer objectives into executable mechanical, electrical, controls, firmware, fabrication, and test work while identifying interface owners, unresolved dependencies, and integration risk.
- Performed hands-on electrical and mechanical integration, embedded configuration and firmware loading, bring-up, troubleshooting, and sea-test support across autonomous platforms.
- Evaluated and recommended embedded-control architectures, compute and I/O partitioning, interface approaches, and component selections by balancing technical capability, integration risk, procurement, schedule, and maintainability.
- Bridged customer needs, discipline specialists, fabrication, procurement, and test personnel so cross-domain decisions are made with the full system rather than any single subsystem in view.

Industry Expert / Independent Consultant (Contract) | Tegus by AlphaSense / *Remote*

2026 – Present

- Selected as a paid industry expert for AlphaSense client research engagements involving engineering, manufacturing, autonomous systems, defense technology, PCB development, and related technical markets.
- Provide practitioner-level technical and market insight by drawing on hands-on experience in electromechanical systems integration, autonomous platforms, electronics development, manufacturing, supplier selection, and prototype-to-field execution.
- Advise clients on engineering decision-making, technology adoption, make-versus-buy considerations, vendor and supplier evaluation, manufacturing constraints, competitive differentiation, and the practical challenges of developing and deploying complex hardware systems.
- Translate complex engineering subjects into concise, decision-useful insights for client research while clearly distinguishing firsthand technical experience from broader industry knowledge, market interpretation, and areas outside direct expertise.
- Participate in structured expert interviews while maintaining strict boundaries around proprietary, confidential, controlled, and non-public information.
- Screened and recommended for client research on the three-to-five-year unmanned-maritime market outlook, major industry participants, Department of Defense priorities, and the engineering realities that differentiate fieldable systems from demonstrations.
- Screened and recommended for research involving PCB design services and AI-assisted PCB tools based on firsthand experience designing 30+ custom boards, selecting external board and assembly houses, and integrating electronics into complete electromechanical systems.

AI Training Expert — Fusion CAM (Contract) | Micro1 / *Remote*

2026 – Present

- Selected as one of the lead beta-testing trainers responsible for completing initial client tasks, identifying workflow deficiencies, and helping develop the processes and guidance used by the broader expert team.
- Generate expert training data by developing Fusion CAM manufacturing strategies while recording real-time reasoning around workholding, setup planning, tool selection, process sequencing, machine constraints, tolerances, verification, alternatives, and recovery from errors.
- Proposed a narrated think-aloud workflow that captures tacit expert judgment, uncertainty, competing approaches, mistakes, corrections, and validation evidence rather than relying solely on post-process demonstrations.
- Evaluate task feasibility, ambiguity, safety, requirement compliance, process correctness, and output quality; provide detailed technical feedback supporting task, rubric, and program refinement.
- Translate manufacturing expertise into repeatable procedures, evaluation criteria, and quality standards for AI training and assessment across a distributed technical team.
- Participate in client discussions as a technical expert, communicating practical limitations, capability gaps, workholding considerations, and opportunities to improve the fidelity of the training program.
- Develop complete multi-setup CAM solutions from source geometry, including stock definition, fixture and workholding strategy, setup orientation, tool selection, roughing and finishing sequences, simulation, and manufacturability review.
- Create evaluation rubrics combining binary acceptance criteria, free-form evaluator guidance, golden-reference checks, and explicit judgment calls so complex expert work can be assessed consistently.
- Diagnose ambiguous or internally inconsistent task states, document the evidence and recovery path, and distinguish true CAM errors from hidden setup, fixture, machine, or tooling constraints.
- Traced an apparent facing and toolpath anomaly to a hidden Op2 fixture, documented the conflicting state, and validated the corrected part rather than treating the visual symptom as a CAM failure.
- Advocate for including workholding design and physical-machining validation in the training workflow because fixture strategy, machine behavior, and recovery from real process errors contain substantial tacit expert knowledge.

Electronics Technician (Engineering), Weapons, Robotics & Controls Eng. Dept. | U.S. Naval Academy / *Annapolis, MD*

2023 – 2026

- Designed and deployed **30+ custom PCBs** for sensor integration, motion control, power distribution, and embedded processing across autonomous marine and robotics platforms.
- Architected and deployed a fleet of **20+ autonomous surface vessels** for swarm and multi-agent research, using a dual–Raspberry Pi control architecture and standardized sensor-payload interfaces; managed integration risk across repeated live-water deployments.
- Designed the electromechanical and PCB architecture for **SENTRY**, an open-source pan-tilt turret platform later released publicly by the institution.
- Integrated sensors, actuators, embedded controllers, communications, and power electronics into field-deployable platforms; improved reliability through iterative field testing, fault isolation, and redundancy planning.
- Supported advanced prototyping, research, and instruction in robotics, controls, embedded systems, autonomous vehicles, PCB development, and digital manufacturing for faculty and cadet projects.
- Owned complete hardware-development cycles from ambiguous research objectives through architecture, CAD, schematic capture, PCB layout, fabrication, assembly, embedded bring-up, troubleshooting, documentation, and field validation.
- Developed reference firmware and a student-writable software interface for SENTRY, proving closed-loop pan-tilt operation while preserving the platform as an instructional controls and autonomy exercise used by 100+ students annually.
- Standardized payload, sensor, communications, power, and mechanical interfaces across autonomous surface vessels to simplify reconfiguration, reduce wiring ambiguity, and support repeated swarm-research deployments.
- Built experimental hardware and conducted the test campaigns behind peer-reviewed research on GNSS flight-controller evaluation and support-free additive manufacturing, contributing as a published co-author.
- Designed and maintained laboratory fixtures, test infrastructure, fabrication workflows, and technical documentation supporting repeatable electronics bring-up, mechanical integration, fault isolation, and student use.

Adjunct Professor — Engineering Design | Anne Arundel Community College / *Arnold, MD*

2024 – Present

- Teach SolidWorks and advanced CAD for mechatronics and manufacturable part development — assemblies, drawings, tolerancing, and design-for-manufacturing.
- Students achieve a **~75% pass rate** on the Certified SolidWorks Associate (CSWA) exam.
- Deliver ENT 260 Solid Modeling as a hands-on engineering-design course covering parametric modeling, assemblies, drawings, design intent, tolerancing, and professional CAD practices.
- Develop and adapt lectures, demonstrations, laboratories, projects, and assessments that connect SolidWorks techniques to real electromechanical packaging, fabrication, and design-for-manufacturing decisions.
- Evaluate student models, assemblies, and drawings for technical correctness, design intent, manufacturability, and compliance with engineering documentation practices.

Mentor, Lead Mentor & Event Manager | FIRST Robotics Competition / *Maryland*

2006 – Present

- FIRST Robotics mentor for **20+ years**; lead mentor for 5, growing a program to 110+ students and mentors across four teams.
- Won the **FIRST Chairman's Award** multiple times — the program's highest honor, recognizing the team that best represents its mission and impact.
- Event-managed **four district competitions and one district championship**, running live technical and logistical operations for thousands of participants under fixed schedules.
- Drove statewide robotics advocacy contributing to **~$250K** in annual Maryland robotics funding through legislative testimony.
- Mentor students and adult volunteers across CAD, mechanical design, machining, additive manufacturing, electronics, wiring, controls, programming, systems integration, troubleshooting, documentation, and technical presentation.
- Served as competition judge, Assistant Event Manager, and Event Manager, bringing both technical-program and live-operations perspectives to team evaluation and event execution.
- Plan and coordinate venue, field, pit, safety, volunteer, schedule, communications, and contingency interfaces for multi-day events, delegating functional ownership while retaining responsibility for cross-team integration and rapid issue resolution.

Engineering Teacher & Program Lead | Various High Schools / *MD / PA*

2012 – 2022

- Designed and built engineering fabrication labs (additive manufacturing, CNC, laser cutting, PCB/electronics prototyping) and the project-based curricula that ran in them.
- Guided student teams through full mechanical/embedded design cycles — CAD, fabrication, assembly, test, and troubleshooting.
- Taught project-based engineering, CAD, robotics, electronics, manufacturing, and PLTW Aerospace Engineering, connecting analytical concepts to the design and fabrication of working hardware.
- Developed original lessons, design challenges, laboratory exercises, rubrics, and assessments spanning requirements, ideation, CAD, prototyping, embedded systems, manufacturing, documentation, testing, and design iteration.
- Established safe, repeatable laboratory operations by selecting equipment and tooling, developing student workflows and safety practices, maintaining fabrication capability, and troubleshooting mechanical, electrical, and computing infrastructure.
- Evaluated student designs, drawings, code, fabricated parts, and integrated systems for technical correctness, manufacturability, documentation quality, and evidence-based iteration rather than completion alone.

SELECTED PROJECTS

- **SENTRY —** Open-source pan-tilt turret platform (originated at USNA, publicly released). Architected and built the complete system — mechanical, PCB, firmware, and enclosure — with reference firmware proving closed-loop operation, then designed the software layer as a student-writable interface. Independently maintain the hardware, PCB, and mechanical design post-departure; used annually by 100+ students, and the subject of an invited Autodesk University 2025 presentation.
- **PID Hover Rig —** Designed a ping-pong-ball levitation demonstrator end to end (mechanical, electrical, and hardware) as a hands-on PID controls teaching platform; students implement the control loop on the hardware as the core exercise.
- **SPARK —** Designing a ruggedized STLINK-V3MODS debug-probe breakout with CAN FD (TCAN1051) and fault / power-good telemetry, standardizing embedded bring-up and eliminating wiring ambiguity; owning the full design lifecycle (schematic, PCB layout, validation).
- **Fusion System Blocks —** Developing an Autodesk Fusion add-in that generates system block diagrams from CAD assemblies — visualizing subsystem boundaries, signal flow, and interfaces to reduce ambiguity in cross-domain design reviews.
- Connector Engineering Field Guide — Developing an open-source Docusaurus reference for electromechanical engineers covering connector-selection workflows, standards and families, datasheet interpretation, MIL-DTL-38999, M8/M12, common design-review failures, and reusable checklists, comparison matrices, cable-drawing guidance, and interface-control-document templates.
- **Home Automation & Digital Twin Platform** — Architected a Home Assistant Yellow-based residential controls and monitoring system integrating HVAC, energy, solar, water, lighting, cameras, presence, network health, and ESP32/ESPHome sensors; building system-health dashboards, watchdogs, predictive-maintenance concepts, and resilient local automations.
- **Nomad Offline Knowledge Node** — Developing a self-contained local knowledge and AI platform on a compact computer with redundant storage, designed to preserve documentation and provide useful search and inference capabilities without cloud connectivity.

PUBLICATIONS & PRESENTATIONS

Built the hardware and ran the experimental testing behind the published data below.

- Co-author — *"GNSS Emulator for Test and Evaluation of Flight Controller Performance,"* ICUAS, May 2025.
- Co-author — *"A Geometry-based Approach for Support-free Additive Manufacturing of Structures with Large Overhang Angles and Closed Features,"* IEEE/RSJ IROS, Oct 2024.
- Selected Presenter — *"From Schematics to Reality: Turbocharging Mechatronics with Fusion,"* Autodesk University, Sept 2025 (based on SENTRY).

EDUCATION

M.Eng., Engineering Management — Old Dominion University · 4.0 GPA (in progress)

M.Ed., Instructional Technology — University of Maryland Global Campus · 4.0 GPA

B.S., Technology & Engineering Education — Millersville University · 3.5 GPA